

Edeviser — Supabase Columns & Data Types Reference

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Enums

Enum Name	Values
assignment_type	assignment, quiz, project, exam
attainment_level	excellent, satisfactory, developing, not_yet
attainment_scope	student_course, course, program, institution
blooms_level	remembering, understanding, applying, analyzing, evaluating, creating
outcome_type	ILO, PLO, CLO
submission_status	submitted, graded
user_role	admin, coordinator, teacher, student, parent
wellness_habit_type	meditation, hydration, exercise, sleep

Tables A–C

academic_calendar_events

Institution-wide calendar events (exam periods, holidays, registration deadlines).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	

semester_id	uuid	Yes	null	FK → semesters.id	
title	text	No	—		Event name
event_type	text	No	—		e.g. exam, holiday, registration
start_date	date	No	—		
end_date	date	No	—		
is_recurring	boolean	No	false		
created_at	timestampz	No	now()		

ai_feedback

Stores AI-generated suggestions and student feedback on them.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
suggestion_type	text	No	—		Type of AI suggestion
suggestion_text	text	No	—		The suggestion content
suggestion_data	jsonb	Yes	null		Structured suggestion metadata
feedback	text	Yes	null		Student's feedback on suggestion
validated_outcome	text	Yes	null		Whether suggestion was helpful
created_at	timestampz	No	now()		

announcements

Course-level announcements posted by teachers/coordinators.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
author_id	uuid	No	—	FK → profiles.id	
title	text	No	—		

content	text	No	—		Rich text body
is_pinned	boolean	No	false		
search_vector	tsvector	No	auto-generated		Full-text search index
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

assignments

Coursework items (assignments, quizzes, projects, exams) with CLO weights and prerequisites.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
created_by	uuid	Yes	null	FK → profiles.id	Teacher who created
title	text	No	—		
description	text	Yes	null		
type	assignment_type	No	'assignment'		Enum: assignment/quiz/project
due_date	timestampz	No	—		
total_marks	integer	No	—		
clo_weights	jsonb	No	'{}'		CLO → weight mapping
prerequisites	jsonb	Yes	null		Prerequisite gate config
is_late_allowed	boolean	No	false		
late_window_hours	integer	No	0		Hours after due date
search_vector	tsvector	No	auto-generated		Full-text search
created_at	timestampz	No	now()		

attendance_records

Per-student attendance for class sessions.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	

session_id	uuid	No	—	FK → class_sessions.id	
student_id	uuid	No	—	FK → profiles.id	
marked_by	uuid	No	—	FK → profiles.id	Teacher who marked
status	text	No	—		present/absent/late/excusec
created_at	timestamptz	No	now()		

audit_logs

Append-only audit trail for admin actions. **Immutable — no UPDATE/DELETE.**

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
actor_id	uuid	No	—	FK → profiles.id	Who performed action
action	text	No	—		e.g. create, update, delete
target_type	text	No	—		Entity type affected
target_id	uuid	Yes	null		Entity ID affected
diff	jsonb	Yes	null		Before/after change diff
ip_address	inet	No	—		Client IP
created_at	timestamptz	No	now()		

badges

Earned badges for students (gamification).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
badge_key	text	No	—		Unique badge identifier
badge_name	text	No	—		Display name
emoji	text	No	''		Badge emoji icon
awarded_at	timestamptz	No	now()		

baseline_attainment

Pre-course CLO attainment scores from baseline assessments.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
course_id	uuid	No	—	FK → courses.id	
clo_id	uuid	No	—	FK → learning_outcomes.id	
score	numeric	No	—		Attainment percentage
correct_count	integer	No	—		
question_count	integer	No	—		
assessment_version	integer	No	1		Reassessment tracking
created_at	timestampz	No	now()		

baseline_test_config

Per-course configuration for baseline assessments.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	One-to-one
time_limit_minutes	integer	No	30		
min_questions_per_clo	integer	No	3		
is_active	boolean	No	true		
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

challenge_participants

Tracks individual/team participation and progress in social challenges.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
challenge_id	uuid	No	—	FK → social_challenges.id	
participant_id	uuid	No	—		Student or team ID

participant_type	text	No	—		'student' or 'team'
current_progress	numeric	No	0		
created_at	timestampz	No	now()		

class_sessions

Individual class meetings within a course section.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
section_id	uuid	No	—	FK → course_sections.id	
session_date	date	No	—		
session_type	text	No	—		lecture/lab/tutorial
topic	text	Yes	null		
created_at	timestampz	No	now()		

course_materials

Learning materials (files, links, videos) within course modules.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
module_id	uuid	No	—	FK → course_modules.id	
title	text	No	—		
description	text	Yes	null		
type	text	No	—		file/link/video/docume
file_path	text	Yes	null		Storage bucket path
content_url	text	Yes	null		External URL
clo_ids	jsonb	Yes	null		Linked CLOs
sort_order	integer	No	0		
is_published	boolean	No	false		
search_vector	tsvector	No	auto-generated		
created_at	timestampz	No	now()		

course_modules

Organizational units within a course (chapters/units).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
title	text	No	—		
description	text	Yes	null		
sort_order	integer	No	0		
is_published	boolean	No	false		
created_at	timestampz	No	now()		

course_sections

Sections within a course, each assigned to a teacher.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
teacher_id	uuid	No	—	FK → profiles.id	
section_code	text	No	—		e.g. "A", "B"
capacity	integer	No	30		
is_active	boolean	No	true		
created_at	timestampz	No	now()		

courses

Academic courses belonging to a program.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
program_id	uuid	No	—	FK → programs.id	
semester_id	uuid	Yes	null	FK → semesters.id	
teacher_id	uuid	Yes	null	FK → profiles.id	Primary teacher
code	text	No	—		e.g. "CS101"

name	text	No	—		
semester	text	No	—		Legacy semester text
academic_year	text	No	—		e.g. "2025-2026"
is_active	boolean	No	true		
search_vector	tsvector	No	auto-generated		
created_at	timestampz	No	now()		

cqi_action_plans

Continuous Quality Improvement action plans for outcome gaps.

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
program_id	uuid	No	—	FK → programs.id	
semester_id	uuid	No	—	FK → semesters.id	
outcome_id	uuid	No	—	FK → learning_outcomes.id	
outcome_type	text	No	—		ILO/PLO/CLO
action_description	text	No	—		Corrective act
responsible_person	text	No	—		
baseline_attainment	numeric	No	—		Before CQI
target_attainment	numeric	No	—		Goal
result_attainment	numeric	Yes	null		After CQI
status	text	No	'planned'		planned/in_pr
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

Tables D–G

departments

Organizational departments within an institution.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	

institution_id	uuid	No	—	FK → institutions.id	
head_of_department_id	uuid	Yes	null	FK → profiles.id	
name	text	No	—		
code	text	No	—		e.g. "CS", "EE"
created_at	timestampz	No	now()		

discussion_replies

Replies within discussion threads.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
thread_id	uuid	No	—	FK → discussion_threads.id	
author_id	uuid	No	—	FK → profiles.id	
content	text	No	—		
is_answer	boolean	No	false		Marked as accepted answer
created_at	timestampz	No	now()		

discussion_threads

Course-level discussion forum threads.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
author_id	uuid	No	—	FK → profiles.id	
title	text	No	—		
content	text	No	—		
is_pinned	boolean	No	false		
is_resolved	boolean	No	false		
created_at	timestampz	No	now()		

evidence

Immutable evidence records linking grades to CLO/PLO/ILO attainment. **Append-only — no UPDATE/DELETE.**

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
submission_id	uuid	No	—	FK → submissions.id	
grade_id	uuid	No	—	FK → grades.id	
clo_id	uuid	No	—	FK → learning_outcomes.id	
plo_id	uuid	No	—	FK → learning_outcomes.id	
ilo_id	uuid	No	—	FK → learning_outcomes.id	
score_percent	numeric	No	—		0–100
attainment_level	attainment_level	No	—		Enum: excellent/sa
created_at	timestampz	No	now()		

fee_payments

Student fee payment records.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
fee_structure_id	uuid	No	—	FK → fee_structures.id	
amount_paid	numeric	No	—		
payment_date	date	No	—		
payment_method	text	Yes	null		
receipt_number	text	Yes	null		
status	text	No	'pending'		pending/completed/re
created_at	timestampz	No	now()		

fee_structures

Fee definitions per program per semester.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
program_id	uuid	No	—	FK → programs.id	
semester_id	uuid	No	—	FK → semesters.id	
fee_type	text	No	—		tuition/lab/library/etc.
amount	numeric	No	—		
currency	text	No	'USD'		
due_date	date	No	—		
created_at	timestamptz	No	now()		

goal_suggestions

AI-generated SMART goal suggestions for students.

Column	Type	Nullable	Default	Key	Not
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
goal_text	text	No	—		
difficulty	text	No	—		easy/medium/hard
week_start	date	No	—		Week the goal targets
status	text	No	'suggested'		suggested/accepted/c
smart_specific	text	Yes	null		SMART breakdown
smart_measurable	text	Yes	null		
smart_achievable	text	Yes	null		
smart_relevant	text	Yes	null		
smart_timebound	text	Yes	null		
cohort_completion_rate	numeric	Yes	null		% of cohort who comp
created_at	timestamptz	No	now()		

grade_categories

Weighted grade categories within a course (e.g., Midterm 30%, Final 40%).

Column	Type	Nullable	Default	Key	Notes
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id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
name	text	No	—		
weight_percent	numeric	No	—		Must sum to 100 per course
sort_order	integer	No	0		
created_at	timestamptz	No	now()		

grades

Grading records for submissions, including rubric selections.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
submission_id	uuid	No	—	FK → submissions.id	One-to-one
graded_by	uuid	No	—	FK → profiles.id	Teacher
total_score	numeric	No	—		Raw score
score_percent	numeric	No	—		0–100
rubric_selections	jsonb	No	'{}'		Per-criterion selections
overall_feedback	text	Yes	null		
graded_at	timestamptz	No	now()		

Tables H–L

habit_tracking

Daily habit completion tracking (legacy — see also `wellness_habit_logs`).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
habit_date	date	No	—		
login	boolean	No	false		
submit	boolean	No	false		

journal	boolean	No	false		
read_content	boolean	No	false		
is_perfect_day	boolean	No	false		All 4 habits done
created_at	timestampz	No	now()		

institution_settings

Per-institution configuration (attainment thresholds, grade scales, etc.).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	One-to-one
attainment_thresholds	jsonb	No	default thresholds		{excellent: 85, satisfactory: 70, developing: 50}
success_threshold	numeric	No	70		% for "meeting expectations"
grade_scales	jsonb	No	'{}'		Custom grade scale config
accreditation_body	text	No	''		e.g. ABET, AACSB
wellness_xp_amount	integer	No	10		XP per wellness habit
created_at	timestampz	No	now()		

institutions

Top-level institution entities. Root of the multi-tenant hierarchy.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
name	text	No	—		
logo_url	text	Yes	null		
accreditation_body	text	Yes	null		
settings	jsonb	No	'{}'		General settings
created_at	timestampz	No	now()		

journal_entries

Student reflective journal entries linked to courses and CLOs.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
course_id	uuid	No	—	FK → courses.id	
clo_id	uuid	Yes	null	FK → learning_outcomes.id	Optional CLO link
content	text	No	—		≥50 words for XP
is_shared	boolean	No	false		Visible to teacher
created_at	timestampz	No	now()		

learning_outcomes

Unified table for ILOs, PLOs, and CLOs (polymorphic via `type` column).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
program_id	uuid	Yes	null	FK → programs.id	Set for PLO/CLO
course_id	uuid	Yes	null	FK → courses.id	Set for CLO only
created_by	uuid	Yes	null	FK → profiles.id	
type	outcome_type	No	—		Enum: ILO/PLO/CLO
title	text	No	—		
description	text	Yes	null		
blooms_level	blooms_level	Yes	null		CLOs only — enum
sort_order	integer	No	0		
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

learning_path_nodes

Ordered assignment nodes in a course learning path with prerequisite gating.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
assignment_id	uuid	No	—	FK → assignments.id	
prerequisite_node_id	uuid	Yes	null	FK → learning_path_nodes.id	Self-referencing
sort_order	integer	No	0		
unlock_threshold	numeric	No	70		% needed on prerequisite
created_at	timestamptz	No	now()		

login_attempts

Rate-limiting table for login attempts (no FK — keyed by email).

Column	Type	Nullable	Default	Key	Notes
email	text	No	—	PK	
attempt_count	integer	No	0		
locked_until	timestamptz	Yes	null		Account lockout time
updated_at	timestamptz	No	now()		

Tables M–O

mastery_recovery_pathways

Tracks student recovery pathways when CLO mastery fails.

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
course_id	uuid	No	—	FK → courses.id	
clo_id	uuid	No	—	FK → learning_outcomes.id	
institution_id	uuid	No	—	FK → institutions.id	

failure_count	integer	No	0		Times failed
status	text	No	'active'		active/failed
activated_at	timestampz	No	now()		
completed_at	timestampz	Yes	null		
expired_at	timestampz	Yes	null		
practice_completed	boolean	No	false		
practice_completed_at	timestampz	Yes	null		
ai_tutor_completed	boolean	No	false		
ai_tutor_completed_at	timestampz	Yes	null		
peer_suggestion_shown	boolean	No	false		
peer_suggestion_applicable	boolean	No	false		
retry_quiz_attempt_id	uuid	Yes	null		
retry_outcome	text	Yes	null		pass/fail
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

micro_assessment_schedule

Scheduled micro-assessments during onboarding (days 2–7).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
assessment_type	text	No	—		
scheduled_day	integer	No	—		Day number (2–7)
scheduled_at	timestampz	No	—		
completed_at	timestampz	Yes	null		
status	text	No	'pending'		pending/completed/dismissed
dismissal_count	integer	No	0		
question_ids	text[]	No	'{'		Array of question UUIDs
created_at	timestampz	No	now()		

notifications

In-app notifications for all users.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
user_id	uuid	No	—	FK → profiles.id	
type	text	No	—		badge/xp/streak/grade/etc.
title	text	No	—		
body	text	Yes	null		
metadata	jsonb	Yes	null		Extra data for deep links
is_read	boolean	No	false		
created_at	timestamptz	No	now()		

onboarding_progress

Tracks student onboarding wizard completion state.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	One-to-one
current_step	text	No	'welcome'		Current wizard step
personality_completed	boolean	No	false		Big Five done
learning_style_completed	boolean	No	false		VARC done
self_efficacy_completed	boolean	No	false		
study_strategy_completed	boolean	No	false		
baseline_completed	boolean	No	false		
day1_completed	boolean	No	false		
baseline_course_ids	text[]	Yes	null		Courses with baseline tests
skipped_sections	text[]	Yes	null		
profile_completeness	numeric	No	0		0–100

assessment_version	integer	No	1		
micro_assessment_day	integer	No	0		
micro_assessment_dismissals	integer	No	0		
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

onboarding_questions

Question bank for onboarding assessments (personality, learning style, baseline, etc.).

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
course_id	uuid	Yes	null	FK → courses.id	For baseline ques
clo_id	uuid	Yes	null	FK → learning_outcomes.id	For baseline ques
assessment_type	text	No	—		personality/learn
dimension	text	Yes	null		e.g. openness, vis
question_text	text	No	—		
options	jsonb	Yes	null		Answer options
correct_option	integer	Yes	null		For baseline only
difficulty_level	text	Yes	null		
weight	numeric	Yes	null		Scoring weight
sort_order	integer	No	0		
is_active	boolean	No	true		
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

onboarding_responses

Student answers to onboarding questions.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	

question_id	uuid	No	—	FK → onboarding_questions.id	
selected_option	integer	No	—		0-indexed option
score_contribution	numeric	Yes	null		Calculated score
assessment_version	integer	No	1		
created_at	timestampz	No	now()		

outcome_attainment

Calculated attainment percentages per outcome per student (UPSERT pattern).

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
outcome_id	uuid	No	—	FK → learning_outcomes.id	
student_id	uuid	Yes	null	FK → profiles.id	Null for a
course_id	uuid	Yes	null	FK → courses.id	
scope	attainment_scope	No	—		Enum: student_c
attainment_percent	numeric	No	0		0–100
sample_count	integer	No	0		Evidence c
last_calculated_at	timestampz	No	now()		

outcome_mappings

Weighted mappings between outcomes (CLO→PLO, PLO→ILO). Many-to-many.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
source_outcome_id	uuid	No	—	FK → learning_outcomes.id	Child (CLO or PLO)
target_outcome_id	uuid	No	—	FK → learning_outcomes.id	Parent (PLO or ILO)
weight	numeric	No	0		Must sum to 100 per child→parent

created_at	timestampz	No	now()		
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Tables P–Q

parent_student_links

Links parent accounts to student accounts.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
parent_id	uuid	No	—	FK → profiles.id	
student_id	uuid	No	—	FK → profiles.id	
relationship	text	No	—		father/mother/guardian
verified	boolean	No	false		Admin-verified link
created_at	timestampz	No	now()		

profiles

User profiles linked to Supabase Auth. Central user table for all roles.

Column	Type	Nullable	Default	Key	Note
id	uuid	No	—	PK	Matches auth.users.id
institution_id	uuid	No	—	FK → institutions.id	
email	text	No	—		
full_name	text	No	—		
role	user_role	No	'student'		Enum: admin/coordinator/teac
avatar_url	text	Yes	null		Storage bucket URL
is_active	boolean	No	true		Soft delete
language_preference	text	No	'en'		i18n
theme_preference	text	No	'light'		light/dark/system
notification_preferences	jsonb	No	'{'}		Per-channel prefs
onboarding_completed	boolean	No	false		
portfolio_public	boolean	No	false		Public portfolio opt-in
tos_accepted_at	timestampz	Yes	null		Terms of Service

last_seen_at	timestampz	Yes	null		Last activity
search_vector	tsvector	No	auto-generated		
created_at	timestampz	No	now()		

program_accreditations

Accreditation records for academic programs.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
program_id	uuid	No	—	FK → programs.id	
accreditation_body	text	No	—		e.g. ABET
status	text	No	'pending'		pending/active/expired
accreditation_date	date	Yes	null		
next_review_date	date	Yes	null		
framework_version	text	Yes	null		
created_at	timestampz	No	now()		

programs

Academic programs (e.g., BSc Computer Science) within an institution.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
department_id	uuid	Yes	null	FK → departments.id	
coordinator_id	uuid	Yes	null	FK → profiles.id	
code	text	No	—		e.g. "BSCS"
name	text	No	—		
description	text	Yes	null		
is_active	boolean	No	true		
created_at	timestampz	No	now()		

question_analytics

Aggregated analytics per question (success rate, discrimination, difficulty calibration).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
question_id	uuid	No	—	FK → question_bank.id	One-to-one
total_attempts	integer	No	0		
correct_count	integer	No	0		
success_rate	numeric	Yes	null		
discrimination_index	numeric	Yes	null		-1 to 1
calibrated_difficulty	numeric	Yes	null		1–5
avg_response_time_seconds	numeric	Yes	null		
quality_flag	text	Yes	null		good/review
last_calculated_at	timestampz	No	now()		

question_bank

AI-generated and manually created questions for adaptive quizzes.

Column	Type	Nullable	Default	Key	
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
course_id	uuid	No	—	FK → courses.id	
clo_id	uuid	No	—	FK → learning_outcomes.id	
created_by	uuid	No	—	FK → profiles.id	
parent_question_id	uuid	Yes	null	FK → question_bank.id	Version change
generation_request_id	uuid	Yes	null		Links to generation request
generation_source	text	No	—		ai/manual
question_type	text	No	—		mcq/true_false
question_text	text	No	—		
options	jsonb	Yes	null		MCQ options
correct_answer	jsonb	No	—		
explanation	text	Yes	null		
explanation_confidence	numeric	Yes	null		0–1 AI confidence

source_chunks	jsonb	Yes	null		RAG source
difficulty_rating	numeric	No	—		1–5
bloom_level	integer	No	—		1–6
labels	text[]	Yes	null		Tags
status	text	No	'draft'		draft/approved
created_at	timestamptz	No	now()		
updated_at	timestamptz	No	now()		

quiz_attempts

Student quiz attempt records with answers and adaptive trajectory.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
quiz_id	uuid	No	—	FK → quizzes.id	
student_id	uuid	No	—	FK → profiles.id	
attempt_number	integer	No	1		
answers	jsonb	No	'{}'		Per-question answers
score	numeric	Yes	null		Null until submitted
started_at	timestamptz	No	now()		
submitted_at	timestamptz	Yes	null		
question_sequence	jsonb	Yes	null		Adaptive question order
difficulty_trajectory	jsonb	Yes	null		Difficulty over time
per_question_times	jsonb	Yes	null		Time per question

quiz_generation_logs

Audit trail for AI question generation requests.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	

institution_id	uuid	No	—	FK → institutions.id	
course_id	uuid	No	—	FK → courses.id	
teacher_id	uuid	No	—	FK → profiles.id	
generation_request_id	uuid	No	—		Unique request ID
model_used	text	No	—		e.g. gpt-4o
question_count_requested	integer	No	—		
question_count_generated	integer	No	—		
chunks_retrieved	integer	No	—		RAG chunks used
prompt_tokens	integer	No	—		
completion_tokens	integer	No	—		
total_tokens	integer	No	—		
latency_ms	integer	No	—		
status	text	No	—		success/error
error_message	text	Yes	null		
created_at	timestamptz	No	now()		

quiz_questions

Questions within a specific quiz instance.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
quiz_id	uuid	No	—	FK → quizzes.id	
question_text	text	No	—		
question_type	text	No	—		mcq/true_false/short_answer
options	jsonb	Yes	null		
correct_answer	jsonb	No	—		
points	integer	No	1		
sort_order	integer	No	0		

quizzes

Quiz definitions with adaptive configuration.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	
title	text	No	—		
description	text	Yes	null		
clo_ids	jsonb	No	'[]'		Targeted CLOs
due_date	timestampz	No	—		
time_limit_minutes	integer	Yes	null		Null = no limit
max_attempts	integer	No	1		
is_published	boolean	No	false		
is_adaptive	boolean	No	false		Uses adaptive engine
adaptation_config	jsonb	Yes	null		Adaptive parameters
created_at	timestampz	No	now()		

Tables R–S

rubric_criteria

Individual criteria within a rubric.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
rubric_id	uuid	No	—	FK → rubrics.id	
criterion_name	text	No	—		
max_points	integer	No	—		
levels	jsonb	No	'[]'		Performance level descriptors
sort_order	integer	No	0		

rubrics

Rubric definitions for grading assignments.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
clo_id	uuid	Yes	null	FK → learning_outcomes.id	
created_by	uuid	Yes	null	FK → profiles.id	
title	text	No	—		
description	text	Yes	null		
is_template	boolean	No	false		Reusable template
created_at	timestampz	No	now()		

semesters

Academic semesters/terms within an institution.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
name	text	No	—		e.g. "Fall 2025"
code	text	No	—		e.g. "F25"
start_date	date	No	—		
end_date	date	No	—		
is_active	boolean	No	false		Only one active per institution
created_at	timestampz	No	now()		

social_challenges

Teacher-created challenges (individual or team-based).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
course_id	uuid	No	—	FK → courses.id	

created_by	uuid	No	—	FK → profiles.id	
title	text	No	—		
description	text	Yes	null		
challenge_type	text	No	—		individual/team
start_date	timestampz	No	—		
end_date	timestampz	No	—		
goal_metric	text	No	—		xp/submissions/streak/e
goal_target	numeric	No	—		Target value
reward_type	text	No	—		xp/badge
reward_value	numeric	No	0		
status	text	No	'draft'		draft/active/completed
notification_sent_90	boolean	No	false		90% progress notificatio
created_at	timestampz	No	now()		

starter_week_sessions

AI-generated starter week plan sessions for new students.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
course_id	uuid	Yes	null	FK → courses.id	
planner_entry_id	uuid	Yes	null		Link to planner
session_type	text	No	—		study/review/practice/etc
description	text	No	—		
suggested_date	date	No	—		
suggested_time_slot	text	No	—		morning/afternoon/evening
duration_minutes	integer	No	—		
status	text	No	'pending'		pending/completed/skipped
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

student_activity_log

Append-only log of student actions for analytics and at-risk detection.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
event_type	text	No	—		page_view/submission/login/etc.
metadata	jsonb	Yes	null		Event-specific data
created_at	timestampz	No	now()		

student_courses

Enrollment junction table (student ↔ course).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
course_id	uuid	No	—	FK → courses.id	
section_id	uuid	Yes	null	FK → course_sections.id	
status	text	No	'enrolled'		enrolled/dropped/completed
enrolled_at	timestampz	No	now()		

student_gamification

Per-student gamification state (XP, level, streak).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	One-to-one
xp_total	integer	No	0		Derived from SUM(xp_transactions)
level	integer	No	1		Calculated from xp_total
streak_count	integer	No	0		Current streak days
streak_longest	integer	No	0		All-time best

streak_freezes_available	integer	No	0		Max 2
last_login_date	date	Yes	null		For streak calculation
leaderboard_anonymous	boolean	No	false		Opt-out of leaderboard
updated_at	timestampz	No	now()		

student_profiles

Extended student profile data from onboarding assessments.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
institution_id	uuid	No	—	FK → institutions.id	
personality_traits	jsonb	Yes	null		Big Five scores
learning_style	jsonb	Yes	null		VARK scores
self_efficacy	jsonb	Yes	null		
study_strategies	jsonb	Yes	null		
profile_completeness	numeric	No	0		0–100
assessment_version	integer	No	1		
completed_at	timestampz	No	now()		
created_at	timestampz	No	now()		

student_wellness_preferences

Per-student wellness habit configuration and preferences.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	One-to-one
enabled_habits	text[]	No	'{}'		Which habits are active
habit_targets	jsonb	No	'{}'		Per-habit daily targets

reminder_times	jsonb	No	'{}'		Per-habit reminder config
parent_visibility	boolean	No	false		Share with parent
dismissed_onboarding_tips	text[]	No	'{}'		
created_at	timestampz	No	now()		
updated_at	timestampz	No	now()		

submissions

Student assignment submissions.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
assignment_id	uuid	No	—	FK → assignments.id	
student_id	uuid	No	—	FK → profiles.id	
text_content	text	Yes	null		Inline text submission
file_url	text	Yes	null		Storage bucket URL
status	submission_status	No	'submitted'		Enum: submitted/graded
is_late	boolean	No	false		
plagiarism_score	numeric	Yes	null		0–100
submitted_at	timestampz	No	now()		

survey_questions

Questions within a survey.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
survey_id	uuid	No	—	FK → surveys.id	
question_text	text	No	—		
question_type	text	No	—		likert/text/mcq

options	jsonb	Yes	null		
sort_order	integer	No	0		

survey_responses

Individual survey responses.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
survey_id	uuid	No	—	FK → surveys.id	
question_id	uuid	No	—	FK → survey_questions.id	
respondent_id	uuid	No	—	FK → profiles.id	
response_value	text	No	—		
created_at	timestampz	No	now()		

surveys

Institution-level surveys (course evaluation, exit survey, etc.).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	No	—	FK → institutions.id	
title	text	No	—		
type	text	No	—		course_evaluation/exit/cu...
target_outcomes	jsonb	No	'[]'		Linked outcomes
is_active	boolean	No	true		
created_at	timestampz	No	now()		

Tables T–X

timetable_slots

Weekly recurring timetable slots for course sections.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
section_id	uuid	No	—	FK → course_sections.id	

day_of_week	integer	No	—		0=Sunday, 6=Saturday
start_time	time	No	—		
end_time	time	No	—		
room	text	Yes	null		
slot_type	text	No	—		lecture/lab/tutorial

verified_explanations

Teacher-verified explanations for question bank items.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
question_id	uuid	No	—	FK → question_bank.id	
institution_id	uuid	No	—	FK → institutions.id	
verified_by	uuid	No	—	FK → profiles.id	Teacher
explanation_text	text	No	—		
source	text	No	—		ai_generated/manual
is_active	boolean	No	true		
created_at	timestamptz	No	now()		
updated_at	timestamptz	No	now()		

wellness_habit_logs

Individual wellness habit completion logs (meditation, hydration, exercise, sleep).

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
wellness_type	wellness_habit_type	No	—		Enum: meditation/hydration/e
date	date	No	—		
value	numeric	Yes	null		Duration/amount
completed_at	timestamptz	No	now()		

created_at	timestampz	No	now()		
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xp_events

Admin-created bonus XP events with time-bounded multipliers.

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
institution_id	uuid	Yes	null	FK → institutions.id	Null = global
name	text	No	—		
description	text	Yes	null		
event_type	text	No	—		
xp_multiplier	numeric	No	1		e.g. 2.0 for double XP
bonus_xp	integer	No	0		Flat bonus
starts_at	timestampz	Yes	null		
ends_at	timestampz	Yes	null		
is_active	boolean	No	true		
created_at	timestampz	No	now()		

xp_transactions

Append-only XP transaction ledger. **Immutable — xp_total is derived from SUM.**

Column	Type	Nullable	Default	Key	Notes
id	uuid	No	gen_random_uuid()	PK	
student_id	uuid	No	—	FK → profiles.id	
team_id	uuid	Yes	null	FK → teams.id	For team XP
source	text	No	—		login/submission/grade/journal/
xp_amount	integer	No	—		Final XP (after multipliers)
base_xp	integer	Yes	null		Before multipliers
final_xp	integer	Yes	null		After multipliers
multipliers	jsonb	No	'{}'		Applied multiplier details

scope	text	No	'individual'		individual/team
reference_id	uuid	Yes	null		Related entity ID
note	text	Yes	null		
created_at	timestamptz	No	now()		

Views

leaderboard_weekly

Materialized view for weekly leaderboard rankings.

Column	Type	Nullable	Notes
student_id	uuid	Yes	FK → profiles.id (via student_gamification)
institution_id	uuid	Yes	FK → institutions.id (via profiles)
full_name	text	Yes	From profiles
xp_total	integer	Yes	From student_gamification
level	integer	Yes	From student_gamification
streak_current	integer	Yes	From student_gamification
global_rank	bigint	Yes	Computed rank

Functions

Function	Arguments	Returns	Purpose
auth_user_role()	none	text	Returns current user's role from JWT
auth_institution_id()	none	uuid	Returns current user's institution_id from JWT
expire_stale_recovery_sessions()	none	integer	Expires old mastery recovery pathways, returns count
get_wellness_aggregate_stats(p_institution_id)	uuid	table(total_logs, unique_students, wellness_type)	Aggregated wellness stats per institution

Total: 57 tables + 1 materialized view + 4 functions + 8 enums

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